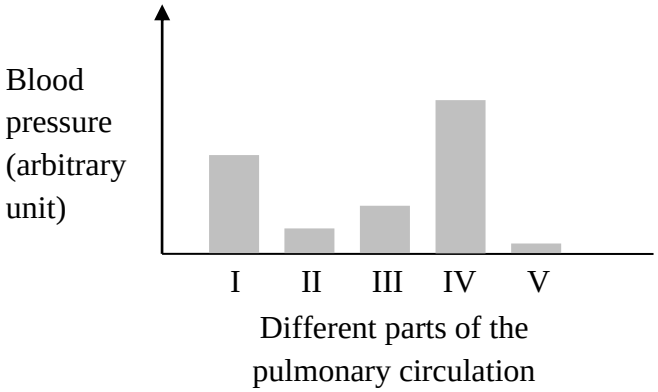
Multiple-choice questions

|!|BCEMA0090000001|!| ★★

The graph below shows the blood pressure of blood vessels at different parts of the pulmonary circulation.



Which of the following correctly shows the direction of blood flow from the heart along the pulmonary circulation?

A. I → II → III → IV → V
B. V → II → III → I → IV
C. IV → III → II → V → I
D. IV → I → III → II → V

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0090000002|!| ★

Which of the following cause(s) the heart ventricles to fill with blood?

(1) contraction of atria
(2) contraction of ventricles
(3) opening of semi-lunar valves at the aorta and pulmonary artery

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000003|!|

What is the state of the atria-ventricular valves (bicuspid and tricuspid valves) and semi-lunar valves when the two ventricles contact?

	Atria-ventricular valves	*Semi-lunar valves*
A.	closed	closed
B.	closed	open
C.	open	open
D.	open	closed

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0090000004|!| ★

Which of the following combinations correctly shows the conditions of the blood traveling through the pulmonary vein?

	Pressure	*Oxygen level*	*Carbon dioxide level*
A.	low	low	high
B.	low	high	low
C.	high	low	high
D.	high	high	low

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0090000005|!| ★

Which of the following combinations is the correct route of the flow of oxygenated blood?

A. left atrium → left ventricle → pulmonary vein → pulmonary artery
B. right atrium → right ventricle → pulmonary artery → pulmonary vein
C. pulmonary vein → left atrium → left ventricle → aorta
D. pulmonary artery → left atrium → left ventricle → aorta

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000006|!|

Which of the following statements about the human circulatory system is ***incorrect***?

A. The aorta and pulmonary veins carry oxygenated blood.

B. The wall of the left ventricle is the thickest among the four heart chambers.

C. The bicuspid and tricuspid valves close when the ventricles contract.

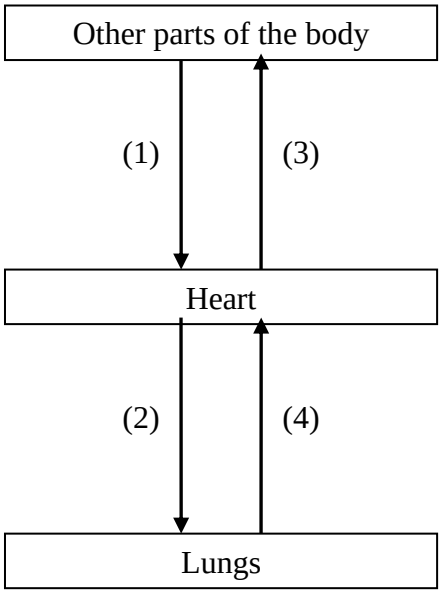
D. The heart tendon contracts to keep the bicuspid and tricuspid valves open.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0090000007|!| ★

The flowchart below shows the blood flow to and from the lungs, heart and other parts of the body. The arrows (1), (2), (3) and (4) represent different blood vessels.



Which two blood vessels carry blood with high oxygen concentration?

A. (1) and (2)

B. (1) and (3)

C. (2) and (3)

D. (3) and (4)

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0090000008|!|

How can phagocytes protect the human body against diseases?

(1) produce antibodies

(2) facilitate the formation of blood clots

(3) engulf and digest invading pathogens

A. (1) only

B. (3) only

C. (1) and (2) only

D. (1) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0090000009|!| ★

Which of the following combinations correctly describes the pulmonary artery?

	Muscle in walls	*Lumen size*	*Blood carried*
A.	thick	small	deoxygenated
B.	thick	small	oxygenated
C.	thin	large	deoxygenated
D.	thin	large	oxygenated

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0090000010|!| ★

When closed, the tricuspid valve

A. prevents blood flowing from left ventricle to left atrium.

B. prevents blood flowing from right ventricle to right atrium.

C. allows blood to flow from left atrium to left ventricle.

D. allows blood to flow from right atrium to right ventricle.

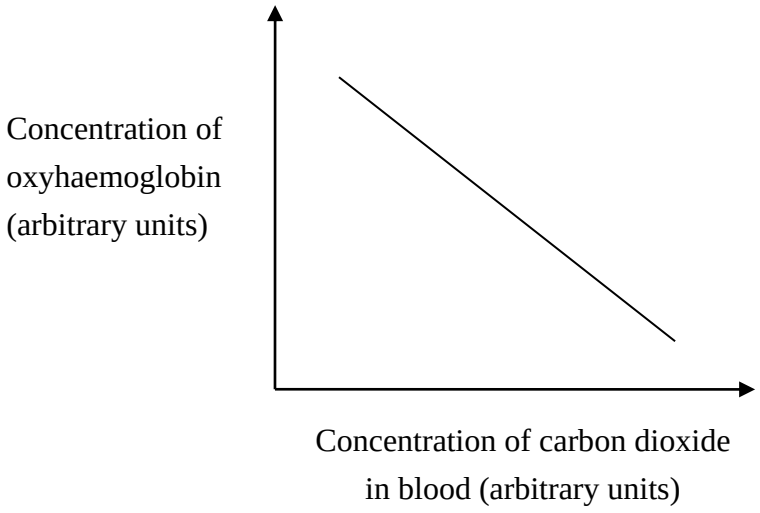
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0090000011|!| ★

The graph below shows the relationship between the concentration of carbon dioxide and oxyhaemoglobin in the blood.



Which of the following is the correct statement describing the relationship shown by the graph?

A. The concentration of oxyhaemoglobin increases with the increase in carbon dioxide concentration.
B. The concentration of oxyhaemoglobin decreases with the increase in carbon dioxide concentration.
C. The concentration of oxyhaemoglobin is directly proportional to the concentration of carbon dioxide.
D. Increasing carbon dioxide concentration has no effect on the concentration of oxyhaemoglobin.

☐

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0090000012|!| ★

Which of the following correctly shows the route taken by blood in making one complete circuit of the body, starting from the leg?

A. leg → liver → heart → lung → leg
B. leg → heart → lung → liver → leg
C. leg → heart → lung → heart → leg
D. leg → lung → heart → lung → leg

☐

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##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0090000013|!| ★★ **Cross topic**

In the capillary network of pancreas, which of the following substances will have a net movement from the pancreatic cells to blood?

(1) carbon dioxide
(2) insulin (a hormone secreted by pancreatic cells)
(3) pancreatic juice

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0090000014|!| ★

Which of the following blood vessels carry deoxygenated blood?

(1) umbilical vein
(2) pulmonary vein
(3) hepatic portal vein
(4) renal vein

A. (3) and (4) only
B. (1), (2) and (3) only
C. (1), (2) and (4) only
D. (2), (3) and (4) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0090000016|!| ★

Which of the following is a correct description of the difference between lymph vessels and veins?

A. Valves are present in veins but not in lymph vessels.
B. Lymph vessels are blind ended while veins are not.
C. Blood flow in veins is maintained by the pumping force of heart while flow of tissue fluid in lymph vessels is maintained by muscle contraction.
D. Veins carry blood towards the heart while lymph vessels carry lymph away from the heart.

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000017|!|

Which of the following blood vessels contains blood that has the highest concentration of oxygen?

A. pulmonary artery
B. dorsal aorta
C. vena cava
D. hepatic portal vein

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000018|!|

Which of the following statements about centrifuged blood is ***incorrect***?

A. The blood is separated into two distinct layers.
B. The upper layer is yellow in colour.
C. Dissolved nutrients can be found in the lower layer.
D. The lower layer is composed of blood cells.

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000019|!|

Which of the following blood components is mainly used for carbon dioxide transport in our bodies?

A. red blood cells
B. white blood cells
C. blood platelets
D. plasma

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000020|!|

Which of the following blood components do ***not*** take part in body defence?

A. red blood cells
B. white blood cells
C. blood platelets
D. antibodies

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000021|!|

Which of the following blood components is/are destroyed in the liver and spleen?

(1) red blood cells
(2) white blood cells
(3) blood platelets

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000022|!|

Which of the following statements about blood vessels is ***incorrect***?

A. Veins are blood vessels that transport deoxygenated blood.
B. Capillaries have smaller lumens than arteries.
C. Arteries are blood vessels which carry blood away from the heart.
D. There are valves at the base of the pulmonary artery and the aorta.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000023|!|

Which of the following statements about the circulatory system is correct?

A. Heart muscle is under voluntary control.
B. The pulmonary artery supplies the heart with oxygenated blood.
C. Inferior vena cava is connected to the left atrium.
D. The coronary veins carry carbon dioxide and other wastes away from the cardiac muscle.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000024|!|

Which of the following statements about the valves is ***incorrect***?

A. Semilunar valves prevent the backflow of blood into the ventricles when the ventricles relax.
B. Tricuspid valve prevents the backflow of blood from the right ventricle into the right atrium.
C. Bicuspid valve is situated between the left atrium and the left ventricle.
D. Semilunar valves are attached to the muscular walls of the ventricles by heart tendons.

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000026|!|

Which of the following statements about the blood vessels and the heart is/are correct?

(1) Arteries are blood vessels that carry oxygenated blood.
(2) Pulmonary vein drains blood into the left atrium.
(3) The left and right atria contract simultaneously.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000027|!|

Which of the following statements about lymph vessels is ***incorrect***?

A. Valves are present in lymph capillaries.
B. The lymph vessels finally open into two large veins near the neck region.
C. The lipids absorbed by lacteals are transported in the lymph in lymph vessels.
D. Contraction of skeletal muscles can drive the lymph to flow along the lymph vessels.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000028|!|

Which of the following body parts can produce blood cells?

(1) bone marrow
(2) lymph nodes
(3) spleen

A. (1) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only

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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000029|!|

Where are old red blood cells broken down?

(1) liver
(2) bone marrow
(3) spleen

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000030|!|

Hepatic portal vein

A. transports oxygenated blood from the small intestines to the liver.
B. supplies nutrients to the small intestines.
C. contains a high concentration of glucose after a starchy meal.
D. contains a high concentration of urea after a meal rich in proteins.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0090000031|!| ★

If the bicuspid valves fail to close fully, which of the following will happen?

A. Some blood will flow back from the left ventricle to the left atrium.
B. Some blood will flow back from the right ventricle to the right atrium.
C. Some blood will flow back from the pulmonary artery to the right ventricle.
D. Some blood will flow back from the aorta to the left ventricle.

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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000032|!|

Septum is responsible for

A. keeping the blood pressure in both ventricles the same.
B. preventing the backflow of blood.
C. preventing oxygenated blood and deoxygenated blood from mixing.
D. separating atria from ventricles.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000033|!|

Which of the following factors determine(s) the rate of blood flow in humans?

(1) the diameter of the lumen of the blood vessels
(2) the intensity of the pumping action of the heart
(3) the oxygen content of the blood

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0090000034|!| ★

Which of the following blood vessels carry oxygenated blood?

(1) umbilical artery
(2) pulmonary vein
(3) renal artery

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000035|!|

Which of the following statements about the systemic circulation are correct?

(1) The left ventricle pumps blood through the aorta to all parts of the body.
(2) Exchange of materials occurs in the capillary beds.
(3) The deoxygenated blood returns to the right atrium of the heart through the venae cavae.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000036|!|

Which of the following comparisons between artery and vein is / are correct?

	Artery	***Vein***
(1)	more elastic fibres in the wall	less elastic fibres in the wall
(2)	deep inside the body	near the body surface
(3)	all carry oxygenated blood	all carry deoxygenated blood

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000037|!|

Which of the following are components of blood plasma?

(1) glucose
(2) haemoglobin
(3) hormones

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000038||!|

Which of the following statements about the functions of blood plasma is ***incorrect***?

A. Plasma helps the distribution of heat throughout the body.
B. Oxygen is mainly transported in plasma.
C. Plasma helps in the transport of insulin from the pancreas to the liver.
D. Urea is transported in plasma to the kidneys for excretion in urine.

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000040|!|

Which of the following statements about the hepatic portal vein are correct?

(1) During fasting, it carries blood with the highest concentration of glucose.
(2) It has blood capillaries at both ends.
(3) It carries blood from the small intestine to the liver.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0090000041|!|

Which of the following comparisons between the white blood cell and mature red blood cell in humans is ***incorrect***?

	White blood cell	***Red blood cell***
A.	can be found in the tissue fluid	cannot be found in the tissue fluid
B.	possess a nucleus	does not possess a nucleus
C.	produced by the lymph nodes	produced by the bone marrow
D.	help in body defence	help in carrying oxygen

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0090000042|!|

Which of the following substances is ***not*** found in the blood plasma?

A. insulin
B. urea
C. haemoglobin
D. carbon dioxide

##C##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000043|!|

Which of following elements is essential for the synthesis of haemoglobin?

A. magnesium
B. calcium
C. iron
D. iodine

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000044|!|

Which of the following comparisons between red blood cell and phagocyte is ***incorrect***?

	Red blood cell	***Phagocyte***
A.	biconcave disc-shaped	irregular-shaped
B.	has no nucleus	has a nucleus
C.	has a higher density in blood	has a lower density in blood
D.	has a shorter life span	has a longer life span

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000045|!|

Which of the following is ***not*** a function of blood?

A. defence against pathogens
B. transport of enzymes
C. distribution of heat around the body
D. transport of lipid from the lymphatic system

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000046|!|

If substances are transported in multicellular animals by diffusion but with no transport system, what would happen?

(1) Cells could not get oxygen quickly enough.
(2) Supply of nutrients to the body cells would be too slow.
(3) Metabolic wastes would tend to accumulate in cells.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000047|!|

Which of the following is the character of all arteries?

A. carry oxygenated blood
B. have a larger lumen and thick wall
C. carry blood away from the heart
D. blood flow maintained by muscle contraction

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000048|!|

Which of the following comparisons between arteries and veins is ***incorrect***?

	Arteries	***Veins***
A.	blood flows more rapidly	blood flows more slowly
B.	with thicker muscular wall	with thinner wall
C.	with larger lumen	with smaller lumen
D.	higher blood pressure	lower blood pressure

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000049|!|

Which of the following statements about capillaries is ***incorrect***?

A. Their walls are one-cell thick.
B. Capillaries have the lowest blood pressure among the blood vessels.
C. Lipids can be forced out from capillaries into the tissue fluid.
D. White blood cells can pass through the walls of capillaries.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000050|!|

A portal vein is a blood vessel which

A. carries blood from an artery to an organ.
B. supplies the liver with oxygenated blood.
C. has capillaries at both ends.
D. carries blood from the small intestine.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0090000052|!| ★

Which of the following comparisons between the blood entering and leaving a contracting skeletal muscle is ***incorrect***?

	Blood entering the muscle	***Blood leaving the muscle***
A.	higher oxygen content	lower oxygen content
B.	lower carbon dioxide content	higher carbon dioxide content
C.	lower rate of blood flow	higher rate of blood flow
D.	higher glucose content	lower glucose content

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0090000055|!|

Which of the following statements about a red blood cell is ***incorrect***?

A. It has a life span of about 120 days.

B. It cannot transport carbon dioxide from body cells to lungs.

C. It does not have a nucleus when mature.

D. It is produced in the bone marrow.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000056|!|

How does the biconcave disc shape of red blood cells help their functioning?

(1) It shortens the distance for the diffusion of nutrients into cells.

(2) It increases the total surface area for diffusion of oxygen.

(3) It makes the red blood cells flow more easily in blood vessels.

A. (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000057|!|

Which of the following blood vessels has the thickest wall?

A. aorta

B. pulmonary artery

C. pulmonary vein

D. superior vena cava

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000058|!|

Which of the following is the function of semilunar valves in the aorta?

A. To prevent the backflow of blood into the right ventricle when the right ventricle relaxes.
B. To prevent the backflow of blood into the left ventricle when the left ventricle relaxes.
C. To force blood to flow in one direction.
D. To prevent the mixing of blood between the aorta and the left ventricle.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0090000059|!| ★

Many passengers find that their feet swell after a long flight (e.g. over 12 hours). Which of the following best explains this phenomenon?

A. The atmospheric pressure is getting lower with the increase in the altitude. The decrease in the atmospheric pressure makes most water in the tissue fluid more difficult to return to the capillaries.
B. In the plane, less oxygen is carried in the blood and so the heart pumps faster. Therefore, the water of the tissue fluid cannot return to the blood as fast as the blood flow.
C. The lack of skeletal muscle contractions reduces the blood flow rate in the veins, leading to the accumulation of blood in the legs.
D. The lack of skeletal muscle contractions reduces the flow rate of lymph in the lymph vessels, leading to the accumulation of excess tissue fluid in the legs.

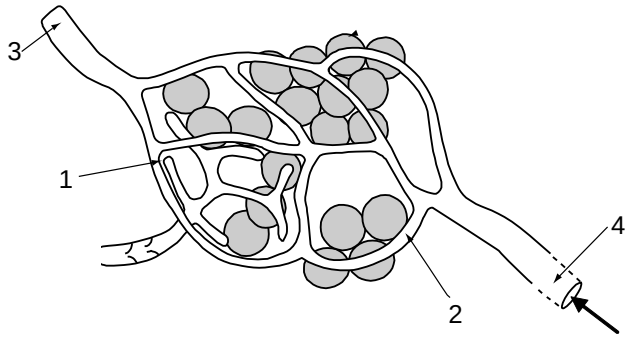
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0090000061|!|

The diagram below shows a certain part of the human circulatory system.



Which of the following sites has the highest blood pressure?

A. 1
B. 2
C. 3
D. 4

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0090000063|!|

Which of the following are the functions of the lymphatic system?

(1) to collect metabolic wastes from the body cells
(2) to transport lipids from the intestinal villi to the blood
(3) to return excess tissue fluid back to the blood

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000064|!|

The function(s) of lymph nodes is/are

(1) to store white blood cells.
(2) to filter out pathogens.
(3) to pump the lymph.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0090000065|!| ★

Which of the following is a correct comparison between lymph vessels and veins?

	Lymph vessels	*Veins*
A.	have no valves	have valves
B.	blind-ended	continuous
C.	contain lipids	contain no lipids
D.	blood flow driven by pumping force of the heart	blood flow driven by contraction of muscles

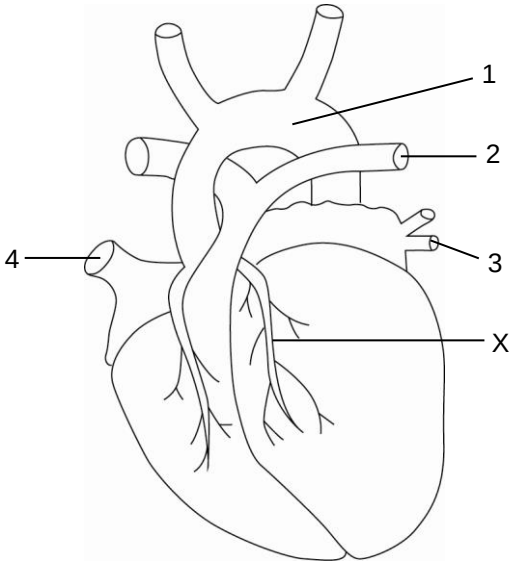
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0090000066|!| ★

Directions: The following two questions refer to the diagram below which shows a human heart.



Under normal condition, which blood vessel contains blood with the highest blood pressure?

A. 1
B. 2
C. 3
D. 4

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0090000067|!| ★

Vessel X is connected to vessel 4. Which of the following about the function of vessel X is correct?

(1) To supply oxygen to the cardiac muscle
(2) To remove carbon dioxide from the cardiac muscle
(3) To transport nutrients directly to the cardiac muscle

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

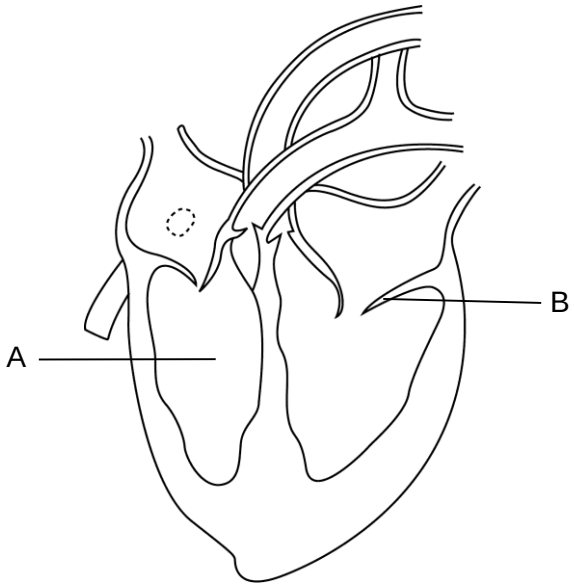
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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0090000068|!|

Directions: The following two questions refer to the diagram below which shows a human heart.



Which of the following combinations correctly matches the structures?

	Structure A	***Structure B***
A.	left ventricle	tricuspid valve
B.	right ventricle	tricuspid valve
C.	left ventricle	bicuspid valve
D.	right ventricle	bicuspid valve

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0090000069|!|

What is the function of structure B?

A. To prevent blood from entering the right atrium when the right ventricle contracts.
B. To prevent blood from entering the left atrium when the left ventricle contracts.
C. To prevent blood from entering the left ventricle when the left atrium contracts.
D. To prevent blood from entering the right ventricle when the right atrium contracts.

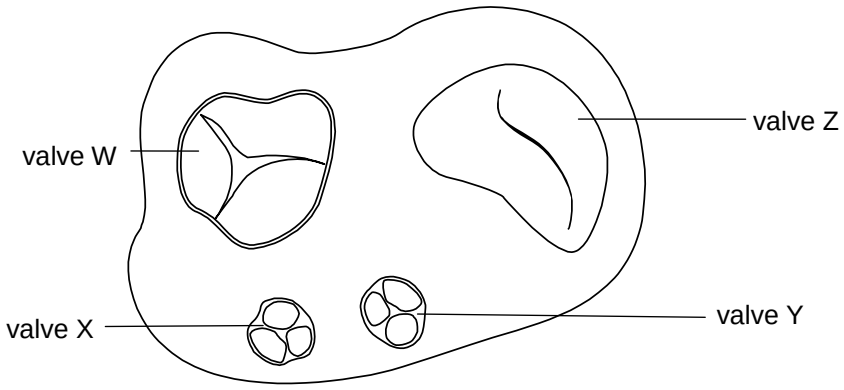
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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0090000070|!|

Directions: The following two questions refer to the diagram below, which shows the cross section of the heart showing the valves.



Which of the following combinations correctly identifies the valves in the figure?

	Tricuspid valves	*Bicuspid valves*
A.	W	Z
B.	W	X
C.	X	Y
D.	X	W

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0090000071|!|

What is the function of valve X and valve Y?

A. To prevent the backflow of blood into the ventricles when the ventricles contract.
B. To prevent the backflow of blood into the ventricles when the ventricles relax.
C. To prevent the backflow of blood into the atria when the atria contract.
D. To prevent the backflow of blood into the atria when the atria relax.

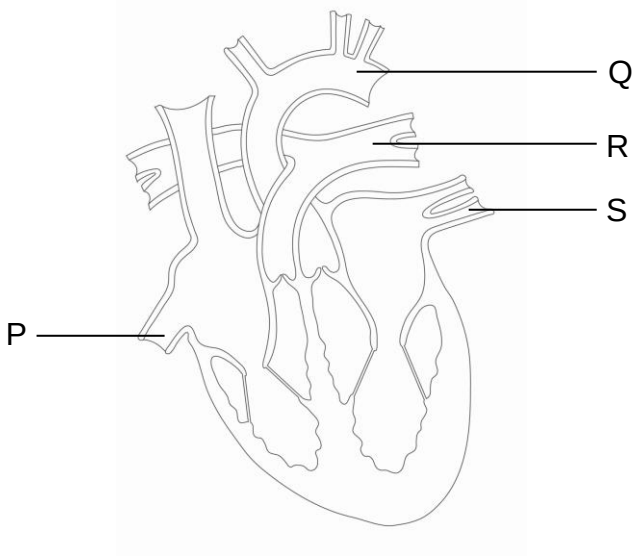
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0090000072|!| ★

Directions: The following three questions refer to the diagram below which shows a human heart and its main blood vessels.



Which of the following vessels carries deoxygenated blood at high blood pressure?

A. P
B. Q
C. R
D. S

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0090000073|!| ★

Which of the following vessels first receives blood with nicotine when a person is smoking?

A. P
B. Q
C. R
D. S

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0090000074|!| ★

Which of the following vessels is the largest artery?

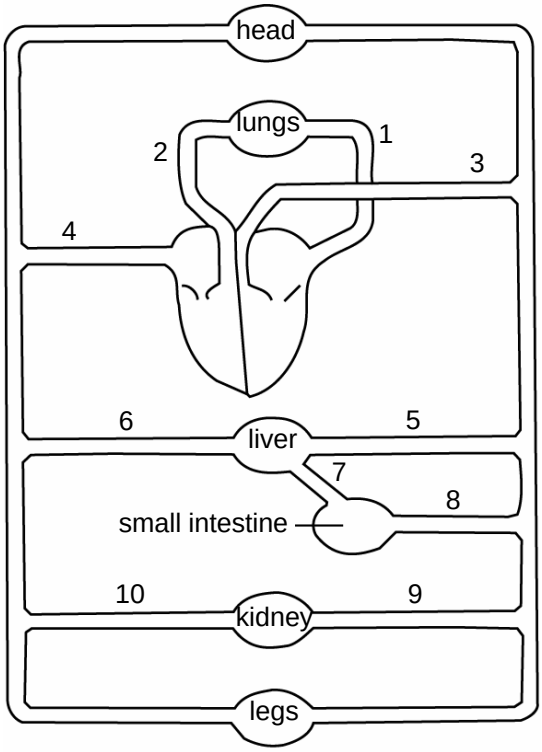
A. P
B. Q
C. R
D. S

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0090000075|!| ★

Directions: The following three questions refer to the diagram below which shows the human circulatory system.



Which of the following blood vessels contains deoxygenated blood?

A. 1
B. 2
C. 3
D. 5

##B##

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Skill assessed	Mark
Analysing data	1

|!|BCEMB0090000076|!| ★

Which of the following blood vessels contains blood with the lowest concentration of urea?

A. 5
B. 6
C. 9
D. 10

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0090000077|!| ★

Which of the following blood vessels contains blood with a high concentration of glucose after a meal?

A. 5
B. 6
C. 7
D. 8

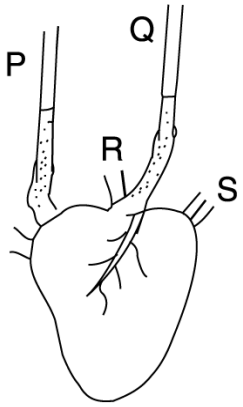
##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMA0090000078|!| ★★ SI

Directions: The following three questions refer to the diagram below which shows an experiment. The vena cava of a pig heart is connected to glass tube P and the pulmonary artery is connected to glass tube Q. A coloured liquid is poured into tube P and come out of the heart into tube Q as shown in the diagram.



If the ventricles are squeezed and then relaxed, what would happen to the level of the coloured liquid in tubes P and Q?

	Tube P	***Tube Q***
A.	remains unchanged	remains unchanged
B.	rises	falls
C.	falls	rises
D.	rises	rises

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0090000079|!| ★★ SI

If the ventricles are squeezed, the coloured liquid will not come out from vessel R. Which of the following is a reason for this?

A. R is a vein which has valves to prevent the backflow of the coloured liquid.
B. R is the aorta which has semilunar valves to prevent the backflow of the coloured liquid.
C. Septum separates the left side and right side of the heart.
D. There are valves located between the atria and the ventricles.

□

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMA0090000080|!| ★★ **SI**

Which of the following shows the correct route of blood flow in the body?

A. P → Q → R → S

B. Q → P → S → R

C. Q → S → R → P

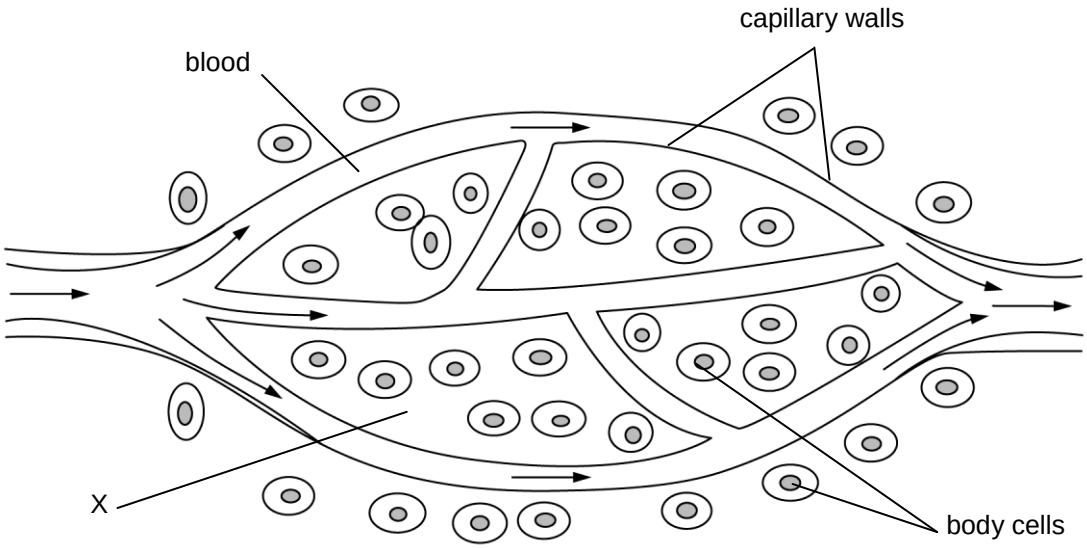
D. R → S → P → Q

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0090000081|!| ★

Directions: The following three questions refer to the diagram below which shows a capillary bed and adjacent cells.



What is the fluid X?

A. blood

B. plasma

C. lymph

D. tissue fluid

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0090000082|!| ★

Which of the following are ***not*** found in fluid X?

(1) carbon dioxide
(2) blood platelets
(3) red blood cells

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0090000083|!| ★

Which of the following statements about fluid X are correct?

(1) It provides a medium for the exchange of materials between the blood and cells.
(2) Excess fluid X will return to the lymph capillaries.
(3) It has the same composition as the blood plasma.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

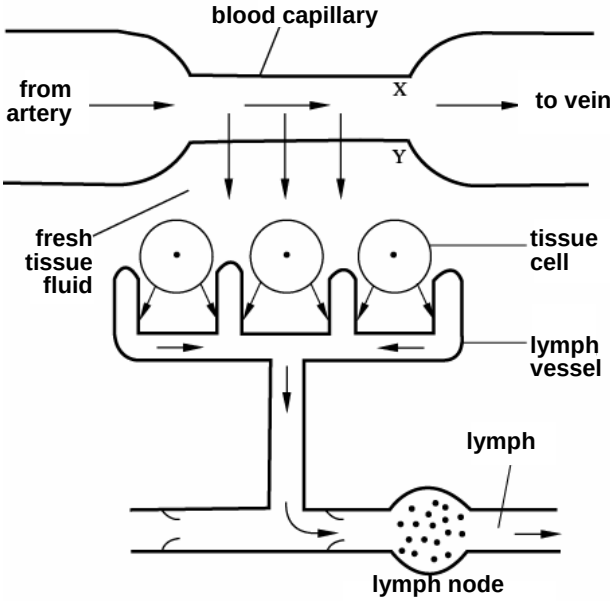
##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0090000084|!|

Directions: The following two questions refer to the diagram below, which shows how the lymph is formed.



Which of the following blood cells ***cannot*** pass from X to Y?

(1) red blood cells
(2) white blood cells
(3) blood platelets

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0090000085|!|

Which of the following comparisons between fresh tissue fluid and lymph is ***incorrect***?

	Fresh tissue fluid	***Lymph***
A.	more urea	less urea
B.	more oxygen	less oxygen
C.	more glucose	less glucose
D.	less carbon dioxide	more carbon dioxide

##A##

Skill assessed	Mark
Understanding basic concepts	1

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